

LODWAR, Kenya

WMO: 636120

Lat: 3.122N

Lon: 35.609E

Elev: 1715

StdP: 13.81

Time Zone: 3.00 (E03)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	69.5	71.3	41.2	40.6	91.2	45.3	47.6	90.1	21.2	83.7	19.0	85.4	2.9	90	0.461

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
3	19.7	99.7	69.5	98.7	69.5	97.8	69.6	76.1	87.0	75.4	86.7	74.7	86.5	11.5	90

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
73.7	133.9	79.3	72.8	129.7	79.3	71.8	125.4	79.5	40.8	87.0	40.1	86.7	39.5	86.1	86.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
23.0	19.7	17.5	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
			54.1	79.4	2.9	2.8	52.0	81.5	50.4	83.1	48.8	84.7	46.7	86.8
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	85.7	84.9	86.6	87.3	86.2	86.1	85.4	84.4	84.8	86.4	86.9	85.4	84.3
	DBStd	2.92	2.61	2.56	3.26	3.91	3.07	2.43	2.17	2.17	2.14	2.66	3.07	2.40
	HDD50	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD65	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDD50	13036	1081	1024	1158	1087	1119	1061	1066	1080	1093	1144	1061	1062
	CDD65	7561	616	604	693	637	654	611	601	615	643	679	611	597
	CDH74	102876	8486	8879	10243	8822	8966	8037	7501	7819	8787	9495	8041	7800
	CDH80	55408	4865	5394	6144	4966	4811	3965	3411	3696	4646	5257	4171	4082
Wind	WSAvg	7.6	6.6	6.3	7.3	7.6	7.8	8.0	7.7	7.9	8.8	8.9	7.7	6.6
Precipitation	PrecAvg	6.6	0.4	0.4	0.9	2.0	1.1	0.3	0.6	0.3	0.2	0.4	0.6	0.4
	PrecMax	12.0	2.7	1.3	3.7	7.4	4.2	2.1	5.2	2.4	2.8	3.5	4.3	2.8
	PrecMin	2.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
	PrecStd	2.8	0.6	0.4	0.9	1.8	1.2	0.6	1.1	0.6	0.6	0.8	0.9	0.7
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	98.9	100.7	100.8	100.0	98.0	95.7	94.5	95.3	97.5	98.1	97.2	97.2
		MCWB	67.9	68.4	70.1	71.4	71.9	71.5	70.4	70.5	69.9	70.4	70.2	68.7
	2%	DB	97.8	99.6	99.7	98.5	96.5	94.6	93.1	93.8	96.1	96.8	95.8	95.9
		MCWB	68.1	68.3	70.0	71.7	71.7	71.0	70.2	70.1	69.8	70.3	70.3	69.1
	5%	DB	96.8	98.7	98.6	97.1	95.3	93.4	91.9	92.9	94.9	95.7	94.8	94.9
		MCWB	68.3	68.3	70.0	71.8	71.7	70.8	70.0	70.0	69.9	70.3	70.4	69.3
	10%	DB	95.6	97.6	97.3	95.6	94.1	92.2	91.0	91.7	93.7	94.7	93.4	93.6
		MCWB	68.2	68.4	70.2	72.0	71.7	70.6	69.9	69.8	69.7	70.3	70.5	69.4
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	75.9	75.3	75.5	77.2	77.0	75.5	74.3	74.7	74.7	75.6	76.2	75.8
		MCDB	91.0	90.0	87.8	86.7	87.2	86.8	85.7	85.7	85.3	85.8	86.7	85.4
	2%	WB	74.3	73.4	74.4	75.9	75.6	74.2	73.1	73.3	73.8	74.5	74.9	74.7
		MCDB	86.9	89.7	88.0	86.4	87.0	86.5	85.0	86.6	86.5	85.8	85.2	85.3
	5%	WB	72.9	72.3	73.6	75.2	74.8	73.4	72.4	72.5	73.0	73.5	74.0	73.8
		MCDB	85.0	90.1	87.8	86.4	86.8	86.9	85.0	86.1	87.3	86.1	84.6	84.4
	10%	WB	71.8	71.3	72.9	74.5	74.1	72.6	71.7	71.7	72.2	72.7	73.3	72.9
		MCDB	86.5	90.3	88.1	86.5	86.6	86.7	85.0	85.5	87.4	86.5	84.3	84.4
Mean Daily Temperature Range	5% DB	MDBR	22.6	22.9	19.7	16.9	15.8	14.6	14.1	14.9	15.8	16.1	16.1	19.3
		MCDBR	24.1	24.1	21.3	18.9	16.7	15.4	14.8	15.9	16.8	16.9	17.7	21.3
		MCWBR	6.7	7.2	5.1	4.7	4.2	4.4	4.1	4.6	4.4	4.1	4.1	5.4
	5% WB	MCDBR	18.1	21.1	18.3	16.5	15.7	14.3	14.3	15.1	15.3	16.3	15.5	16.2
		MCWBR	5.8	7.2	5.2	5.2	4.7	4.7	4.4	5.0	4.4	4.5	5.0	4.9
Clear-Sky Solar Irradiance	taub	0.405	0.437	0.449	0.423	0.432	0.500	0.500	0.454	0.426	0.409	0.414	0.410	
	taud	2.326	2.222	2.212	2.336	2.306	2.063	2.058	2.208	2.285	2.358	2.348	2.335	
	Ebn at Noon	291	284	279	281	271	249	250	268	281	288	286	288	
	Edh at Noon	42	47	48	41	41	52	52	46	44	41	41	41	
All-Sky Solar Radiation	RadAvg	2077	2148	2114	1949	1992	1930	1906	2030	2164	2025	1888	1953	
	RadStd	102	98	131	135	87	95	90	88	87	83	110	118	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (0 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page